

COMPUTER SCIENCE & BUSINESS STUDENT

ANISH ACHAR

TRINITY COLLEGE DUBLIN · **ANISHACHAR.COM**

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EDUCATION

COMPUTER SCIENCE & BUSINESS
Trinity College Dublin
Sep 2026 – Present | Year 1

LEAVING CERTIFICATE
The Institute of Education
Dublin | 2026
Incl. Design & Communication Graphics

SKILLS

- SolidWorks (parts, assemblies)
- SOLIDWORKS API (C#)
- Technical drawing (DCG)
- Flight & flutter analysis
- Mechanical diagnostics
- Drivetrain & brake servicing
- Python (fluent)
- JavaScript (basic)
- Pine Script (TradingView)
- Backtesting & optimisation
- AI-assisted development
- Figma & UI prototyping
- Agile / Scrum & QA testing

COURSES

- CS50's Introduction to Programming with Python
  Harvard University
- Discoveries and Inventions (Robotics)
  CTY Ireland

LANGUAGES

- English | Fluent
- Kannada | Fluent
- Hindi | Fluent

PROFILE

First-year Computer Science & Business student working across **software and hardware**. Software: data and the patterns within it, from Python analysis to optimising a strategy over thousands of backtests. Hardware: seeing ideas come to life, from parametric SolidWorks assemblies to hands-on mechanical builds. Interactive 3D models, charts and live demos of every project below are at **anishachar.com**.

PROJECTS

V8 ENGINE · Parametric CAD assembly — SolidWorks · C# API
- A 5.7 L, 90° cross-plane V8 based on GM's LS: 18 unique parts, 40 in the assembly, generated from one spec sheet by a C# program driving the **SOLIDWORKS API**.
- Rods and pistons are mated to the crank, and piston travel matches the slider-crank equation through a full turn with **zero interference**.

ROCKET FIN CAN · Design study — Python · SolidWorks
- Fin can for a K-class student rocket (K700 motor, 2 km target): wrote a 2-DOF **flight simulation**, Barrowman stability, NACA TN 4197 flutter and structural checks in Python.
- Three iterations gave **2,000 m apogee**, a 1.95× flutter margin and a minimum factor of safety of 2.31; the SolidWorks assembly is generated from the same numbers.

NO CANDLE STRATEGY · Algorithmic trading — Pine Script · TradingView
- Wrote a price-action strategy in Pine Script and tuned it with **11,142 backtests** (sweeps, coordinate descent, grid search) on four FX pairs over 10 months of 15-min data.
- Took it from 0/4 to **4/4 profitable pairs** (profit factor 1.64–1.96, max drawdown under 2.6%); at 20× leverage all four beat the S&P 500's +13.1% in the same period.

GHOST BALL · Pool shot app — JavaScript · Computer vision
- Phone web app that reads a pool table from **one top-down photo**: detects every ball, corrects perspective, then ranks straight, bank, kick and combination pots and draws the best shot.

ZETO · Notion hackathon — Lovable · Notion · Team of 5
- Internship tracker for biomedical and engineering placements in Ireland with **two-way Notion sync**; I built the demo page and made the launch video.

STUDY CONTENT · @lil_indian_boi — TikTok · SimpleStudy
- Documented my Leaving Cert preparation on TikTok (**31K+ views** on my top three videos), which led to a brand ambassador role with SimpleStudy.

PRODUCT DESIGN · CAD prototypes — SolidWorks · DCG
- An **e-bike conversion kit** that mounts a motor and battery on an existing frame, and a contemporary **soap dispenser** taken from sketches to technical drawings for Leaving Cert DCG.

EXPERIENCE

CO-FOUNDER · Resonance, Dublin — Sep 2026 – Present
- Co-founded a web start-up that rebuilds outdated websites for small businesses; building **custom timetabling software** for our first client, VS Driving School.

COMPUTER LITERACY VOLUNTEER · India — Dec 2025
- Taught children **core computer skills** and safe, practical use of technology.

VOLUNTEER BICYCLE MECHANIC · The Bike Hub, Dún Laoghaire — Jun 2025
- Built and repaired **around ten bikes** from scratch under an experienced mechanic, fixing drivetrain and brake faults; all were donated to people in need.
- Favourite job: a **full drivetrain replacement** on a badly corroded bike, from the derailleurs and chain through to the cranks and pedals.

ENGINEERING WORK EXPERIENCE · Tata Consultancy Services — Feb – Mar 2024
- In a small team, reimagined food delivery as a drone-based system; I designed the **app prototype** and user journey in Figma.

TECHNOLOGY WORK EXPERIENCE · Test Triangle, Dublin — Nov 2023
- Shadowed a software tester in an Agile / Scrum team, running functional and usability tests; some of my proposed **improvements were implemented**.